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GeneForgeLang: A Symbolic Language for Genetic Reasoning, Editing, and Simulation

Manuel Menéndez González [1]

1 Fundación de Neurociencias, Spain

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Software

- Review
- Repository
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Summary

GeneForgeLang (GFL) is a symbolic language designed to represent, analyze, and simulate biomolecular processes with unprecedented clarity, granularity, and logical reasoning. GFL allows scientists and AI models to interact through a shared grammar that can describe genetic modifications, transcriptional regulation, protein folding, therapeutic edits, and pathway simulation.

Originally conceived to address the limitations of existing descriptive formats (e.g., FASTA, SBOL, ProForma) and programming languages used in synthetic biology, GFL enables representation of **DNA, RNA, and protein constructs**, **mutation provenance**, **therapeutic delivery**, and **feedback systems**. Its formal grammar and parser support AI-enhanced interpretation, bidirectional translation from scientific literature, and automatic simulation of molecular logic.

Although it is currently used in the context of precision medicine and neurogenetics, GFL is fully extensible to domains like **crop engineering**, **antibiotic resistance**, **pandemic vaccine design**, and **synthetic metabolic pathways**.

Statement of need

With the rise of CRISPR, base editors, and programmable RNA therapies, biomedical science has entered an era of **complex multilayered interventions**. Yet, most researchers still describe their designs through prose, diagrams, or Python scripts—lacking a **semantic, symbolic, and machine-interpretable layer**.

GFL solves this gap by providing a **compact, modular, and logically expressive language** capable of:

- Describing point mutations and large edits with provenance and intent
- Representing epigenetic states and post-translational modifications
- Encoding delivery routes, dosages, and tissue-specific expression
- Enabling feedback loops, probabilistic reasoning, and pathway logic
- Being parsed, validated, and simulated by LLMs and bioinformatics pipelines

Related work

GFL builds upon conceptual and technical precedents:

- SBOL (Galdzicki et al., 2014) provides semantic interoperability but lacks reasoning syntax
- GenoCAD (Czar et al., 2009) enables modular design but not dynamic or clinical modeling
- ProForma (LeDuc et al., 2022) formalizes proteoforms but is protein-only and lacks logic
- Eugene (Densmore et al., 2009) focuses on device constraints, not biological complexity

GFL merges these paradigms into a **unified language for describing biological reality**, particularly suited for AI interaction, therapeutic prototyping, and knowledge extraction.

Software description

GFL includes the following components:

Grammar (`grammar.md`)

A formal specification of the language, defining symbols for:

- Structural abstraction: linear, secondary, tertiary, quaternary (`~`, `:`, `^`, `*`)
- Molecular types: DNA, RNA, protein (`d`, `r`, `p`)
- Logic: conditionals, hypotheses, effects, simulation
- Entities: mutations, edits, delivery vectors, omics pathways

Syntax (`syntax.md`)

An EBNF-based definition of GFL syntax, ensuring:

- Valid sequence of symbolic elements
- Metadata handling via `{}` blocks
- Nested expressions for control logic
- Compatibility with Markdown and plaintext editors

Parser (`parser.py`)

A Python script that:

- Validates GFL statements
- Parses into structured JSON
- Detects semantic errors (e.g., transcription of a protein)
- Supports conversion to SBOL, FASTA, ProForma

GitHub repository

All components and examples are hosted under MIT license at:
https://github.com/Fundacion-de-Neurociencias/GeneForgeLang

Use cases and examples

1. Prime Editing for CFTR F508del (Cystic Fibrosis)

```
~d:CFTR[MUT:SOM:DEL:F508@codon508]
EDIT:Prime(INS:TTC@exon10){pegRNA=CFTR_PE, target=CFTR}
DELIV(mRNA+LNP@IV){target_cells=Pulmonary_Epithelium}
TIME(0d):DELIV(...)
TIME(7d):EFFECT(↑CFTR_protein@apical_membrane)
```

2. miR-21-mediated Drug Resistance in Breast Cancer

```
PATHWAY: ↑miR-21 → INHIBIT(miR-21 → ~r:PTEN_mRNA) → ↓^p:PTEN → ↑^p:AKT
EDIT:Anti-miRNA(miR-21_ASO){delivery=intratumoral}
SIMULATE: {Drug=Doxorubicin, AntagomiR} OUTCOME: ↑Apoptosis
```

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3. CAR-T Design for HER2+ Tumors

```
~d:CAR_HER2 - ^p:Dom(scFv_anti-HER2) - ^p:Dom(CD28_costim) - ^p:Dom(CD3zeta)
TRANSLATE(~r:CAR_HER2_mRNA → ^p:CAR_HER2_protein){cell=T_cell}
INTERACT(CAR_HER2 / HER2){via=immunological_synapse}
EFFECT(↑Tumor_Lysis, ↑T_cell_Persistence)
```


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